

BARRE YASWANTH SATYA NAGA SAI

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PROFESSIONAL SUMMARY

Data Engineer with 4 years of experience designing and delivering Python-based ETL pipelines, cloud data warehousing, data migration, and financial calculations for fintech platforms. Proven contributor to SQL-driven batch data workflows with strong data validation and quality checks, and hands-on experience supporting data science and machine learning pipelines, including feature engineering, model inference, and automation. Track record of delivering accurate, scalable, and business-critical data solutions using AWS S3, Snowflake, and Parquet-based data lakes.

TECHNICAL SKILLS

Programming Languages: Python, C, C++

Data Engineering & ETL: Data Extraction, Data Parsing, Data Migration, ETL Pipelines, Feature Engineering

Database Management: MSSQL, MySQL, SQLAlchemy, Stored Procedures

Cloud & Data Warehousing: AWS S3, Snowflake, Parquet File Format

Data Processing & Automation: Pandas, NumPy, NumPy-Financial, PyPDF2

Web Scraping & OCR: BeautifulSoup, Selenium (Headless Browser), pytesseract

API & Backend Development: FastAPI, Flask, JWT Authentication

Scripting & Tools: Batch Scripting, PyInstaller

DevOps & Deployment: Git, GitHub, GitLab, Docker

Python Frameworks: Tkinter (GUI Development)

EDUCATIONAL QUALIFICATIONS

B.Tech in Computer Science and Engineering (CSE) — Raghu Institute of Technology, JNTU-Kakinada

2018 - 2022 | GPA: 8.14

WORK EXPERIENCE

Software Engineer (Data Engineering) - Assistant Manager | June 2025 - Present

Arthan Finance Pvt. Ltd. - Product-Based Fintech Company

- Designed and implemented Python-based data pipelines to extract data from relational databases using SQLAlchemy, perform data cleaning and feature engineering, and persist processed datasets in Parquet format on AWS S3 for downstream analytics and model training.
- Loaded and migrated processed datasets from AWS S3 (Parquet format) into Snowflake, enabling centralized cloud data warehousing and faster, more reliable access for analytics and reporting.
- Built and maintained machine learning data workflows supporting credit-risk models developed using XGBoost and LightGBM, including model retraining triggers based on performance degradation, following MLOps-oriented practices.
- Executed end-to-end data migration from Loan Management System (LMS) to Loan Origination System (LOS), including data mapping and reconciliation where account identifiers differed across systems.
- Implemented complex financial and credit calculations such as FOIR, LTV, ROI, EMI (PMT), margin PD, monthly expenses, turnover, net surplus, and related metrics using NumPy, NumPy-Financial, Pandas, and SQL, updating customer-level and loan-level tables with high accuracy.
- Developed scheduled and on-demand data processing services using FastAPI and APScheduler, enabling time-based and manual execution of business workflows; containerized and deployed services using Docker.
- Designed an 8-stage customer risk and eligibility scoring framework (RAG score) leveraging historical loan performance indicators such as DPD, CIBIL score, write-off amount, and settlement history, implemented via FastAPI services and database stored procedures.
- Built an Aadhaar document masking solution to anonymize sensitive information in rotated PDF/image inputs using OCR (pytesseract) and Verhoeff checksum validation, ensuring compliance with data-privacy requirements.

- Developed headless web scraping pipelines to collect external datasets, perform data cleaning, and store structured outputs in CSV/XLSX formats, with conditional ingestion into relational databases based on business needs.
- Implemented data crawling and lookup processes to retrieve loan and customer details across multiple tables using identifiers such as PAN, Aadhaar ID, and phone number.
- Optimized SQL queries and Python data access patterns to reduce database load by eliminating row-level loops and batch calls, significantly improving performance and reliability.
- Built robust bulk data ingestion pipelines for CSV and Excel files, including validation, deduplication, and matched-row processing before database insertion.
- Implemented UPSERT (insert/update) mechanisms based on loan_id and customer_id to ensure idempotent data loads and accurate record updates.
- Developed batch prediction pipelines that load trained models using pickle and encoder artifacts, generate inference results, and persist model outputs back into database tables.
- Implemented JWT-based authentication services using Flask APIs to secure internal applications and data services.

Software Engineer | November 2022 - June 2025

NISC Export Services Pvt. Ltd. (NES), Hyderabad

- Developed and maintained Python-based automation scripts for data parsing, transformation, and database updates, enhancing efficiency.
- Designed and implemented custom system applications for database migration (MSSQL to MySQL), data mining, and automation using Selenium, reducing manual effort.
- Utilized Python's Tkinter and PyInstaller for seamless deployment and user-friendly execution of automation tools.
- Built OCR-based applications to extract text and update highlighted coordinates for image processing and document analysis.
- Optimized database queries for faster data retrieval and integrated threading into automation workflows, improving performance and responsiveness.
- Created interactive data visualization dashboards in Power BI to generate actionable insights and track key business metrics.
- Developed web scraping and data mining tools using Selenium and BeautifulSoup to extract, process, and analyze external data sources.
- Collaborated with cross-functional teams to streamline data processing workflows, enhancing efficiency, accuracy, and system performance.

CERTIFICATIONS & PROFESSIONAL TRAINING

- NPTEL: Problem Solving through C Programming
- AWS: AWS Cloud Practitioner Essentials
- Coursera (deeplearning.ai): Introduction to Data Engineering; Source Systems, Data Ingestion & Pipelines; Data Storage & Queries
- Udemy: Data Structures & GUI Development in Python, Web Scraping, Batch Scripting (CMD), MSSQL, Git & GitHub, Power BI